Design notes

DIY CNC epoxy-granite mill · October 2026

Fixed-gantry, epoxy-granite desktop mill sized like the Shariff DMC2 Mini (180 x 300 x 140 mm travel), with a 2020/ACM enclosure mounted on the base and a clear lift-up door.

This document explains the design: the frame, the coolant system, the stiffness study and the cost. The repository layout and the rebuild commands are in the top-level `README.md`.

Frame

Five epoxy-granite castings that bolt together, each with cast-in steel:

Module	Steel inside	Joins to
Base front	2x 2x2x1/8 in tubes (12 in) under the table rails, hollow, sockets for the split spigots, 2x M12 draw rods, the coolant outlet, O-ring groove on the split face	Rear half: 1-1/2 in spigots slide inside the sockets (printed sleeves take up the clearance); the draw rods pull it tight from pockets in the front face
Base rear	2x 2x2x1/8 in tubes (12 in), 2x 1-1/2 in spigots (6 in), 4 printed socket liners, M12 coupling nuts	-
Upright (x2), a raked buttress	2x 1-1/2x1-1/2x1/8 in posts (12 in) that stick out the bottom, 3/8 in top plate, O-ring groove under the foot	Posts drop into the lined sockets, a dowel locates it, 4x M12 tie rods pull it down (two from pockets at the top, two from pockets in the sloped front)
Beam (140 deep x 140 high)	2x 2x2x1/8 in tubes (18 in), drilled 14 mm where the inner cap screws pass	A dowel and 4x M10 cap screws per side into the upright top plates

Nothing you buy gets cut. The rails (including the 600 mm Y pair), ballscrews and tube are used at the lengths they're sold in: tube in 18, 12 and 6 in pieces of two US sizes (remnants are fine, lengths +/- 1/4 in, 0.120 in wall works as well as 1/8 in). The risers, aluminium plates and the upright top plates are ordered cut to size. The only cutting left is the M12 threaded rod. The only machining is drilling the beam tubes (4 holes each) and drilling and tapping the risers, top plates and support pads. Use plain steel, not galvanized: epoxy bonds poorly to zinc. Wire-brush off rust and loose scale and degrease just before casting; paint only the part of each leg that hangs below the base.

Keeping the epoxy down. About 10.6 kg of epoxy (8.5 % of the 125 kg the castings would weigh with sand; 135 kg with the alumina):

- The base sockets, posts and legs stay hollow. Printed plugs close their ends so no mix gets in; fill them with dry sand first if you want extra damping. The beam tubes are filled, because the cap screws pass through them.
- The base's underside pockets follow the stiffness study below: two full-depth girders (80 mm wide) under the table rails, a full-depth tie between them just behind the split, solid knees from the girders out to the upright sockets, and a 25 mm top skin almost everywhere else. The 70 mm front wall carries the coolant trough. Every pocket is open to the bottom, so the cores (printed shells or foam blocks) pull straight out; a few are shallow steps (40 or 65 mm deep) under the trough band, the draw rods, the knees and outboard of the sockets (`base_pockets` in `model/build_cnc.py`).
- The sockets are printed PETG liners cast into the base, not steel tube.

- The mix matters as much as the geometry. Use a well-graded aggregate (gravel up to about 10 mm, sand, fine sand, and whiting, i.e. calcium carbonate powder) and vibrate it. The split in the build guide follows a modified Fuller curve, P(d) = (d^q - d_min^q) / (D^q - d_min^q) with D = 10 mm and q about 0.45: q = 0.5 (classic Fuller) is a little coarse and needs more epoxy to wet, q = 0.37 (Andreasen) packs tighter but its extra powder soaks up epoxy.
- **Aluminium oxide in the beam and uprights.** Alumina grit (E about 370 GPa for brown fused, a little less for black, against about 70 for quartz and granite) replaces the sand, volume for volume, in those three castings only (`analysis/alumina.py`): brown #16 for the coarse sand, black 70 grit for the fine sand. A differential-effective-medium estimate of the mix (the plain mix comes out at 34 GPa, close to the 30 GPa the frame model uses) puts it 22 % stiffer with the #16 alone, 10 % with the 70 grit alone and 37 % with both. On the frame, both come to about +17 / +16 / +21 % in X / Y / Z for $150 (a 50 lb and a 25 lb bag); the #16 alone is about +10 %. Doing the base as well would take about 60 kg more grit for another 8-11 %. It's an estimate, so the build guide's tap-tested bars check it, and sand is the fallback. Other options: **Graded industrial quartz** (Agsco and similar) is clean, dry and consistent but costs more and ships heavy; pool filter sand is the local equivalent. **Ceramic microspheres** (Zeeospheres) help the mix flow at low epoxy, but whiting does most of that for far less. **Epoxy grout** (Laticrete SpectraLock) is fine for a test bar or a repair, but it's fine-grained (about 15-20 % binder) and far too expensive for 125 kg. Whiting is used instead of silica flour because it has no crystalline silica: a plain dust mask is enough. A good mix gets to 8.5 % epoxy, which fits a single 3 gal kit (10 % needs a second, small kit); the build guide's jar test and test bars check it before anything is cast.

What the rails and supports bolt to. Nothing bolts to the cast-in tubes; they stiffen the castings and form the joints. The rails screw into steel risers, and the risers are bedded on the castings with a thin epoxy grout and held down by counterbored M5 screws (under the rail, between its own screws) into cast-in inserts: 5 per riser on the base (Y), 4 on the beam (X). The Y ballscrew supports (BK12/BF12) bolt into M6 inserts in the base top. The X supports (BK10/BF10) sit over the front beam tube, where an insert won't fit, so they bolt to two 1/4 in steel pads cast flush into the beam face and bearing on the tube. The Z axis mounts on the aluminium plates and needs no inserts. See the build guide.

The enclosure's 2020 frame bolts to the base top with angle brackets into cast-in M5 inserts, so the base is the enclosure floor. Seal the top with an epoxy gel coat; coolant runs to the two drains at the front. The door laps over the base front face.

Assembly: base halves (spigots in, tighten the draw rods), uprights (posts into sockets, tighten the tie rods), beam (cap screws), axes, enclosure. To break it down, take the X and Y motors off first (they pass through the enclosure panels), then reverse the order. Heaviest piece: the rear base half (about 42 kg). The upright feet run forward across the base split, so the base halves have to be joined before the uprights go on.

Stiffness study (`analysis/`)

A finite-element model of the frame was used to see where the epoxy granite does the most good, and a topology optimization was run on the base.

Model. The castings and cast-in steel are voxelized from this model at 10 mm (66k brick elements). Each voxel's stiffness is its mix of epoxy granite (E = 30 GPa) and steel; the joints between castings are treated as bonded, which is what the preload is for. A 1 kN force on the tool goes into the beam through the X blocks (the X component through the X ballscrew to its BK bearing) and, reversed, into the base through the Y blocks (the Y component through the Y screw to its BK). The result is the frame's own stiffness at the tool, tool against work, for X at the centre and end, the table at both ends and the centre, and the tool 50 mm above the table. Rails, blocks, screws and the spindle are not in it; the screws alone are roughly 30-50 N/um, so the frame should stay well above that.

Findings (frame only, N/um at the tool, mean over the load cases):

Frame	Epoxy granite	X	Y	Z
Previous design	42.4 L	48.9	80.7	57.9
Same, base made rigid		87.2	130.9	80.2
Same, uprights rigid		83.3	103.9	102.6
Same, beam rigid		122.9	175.7	125.9
Base with no pockets at all (+14 L)	56.8 L	61.5	90.1	67.0
Beam 120 -> 140 mm deep only (+1.5 L)	43.9 L	51.1	90.9	63.8
Optimized base + 140 mm beam	43.7 L	54.0	94.5	68.0
+ riser hold-downs, support pads, uncut 600 mm Y rails	43.7 L	55.7	96.8	68.9
Base + uprights optimized together, same material (free-form uprights)	43.7 L	63.9	103.3	85.2
Raked-buttress uprights, base re-laid around them	52.6 L	77.5	107.5	93.0
+ blunt upright toe, rounded edges, deeper slope pockets	53.0 L	78.2	110.1	95.5
Now: + alumina grit in the beam and uprights (estimate)	**53.0 L**	**91.3**	**127.3**	**115.7**

- The beam is the biggest share of the deflection, then the base and the uprights about equally. Litre for litre, epoxy in the beam is worth roughly ten times as much as epoxy in the base, so the beam now fills the upright tops (140 deep instead of 120).
- In the new frame the beam is still the biggest share. Making the uprights rigid would add about 75 / 25 / 85 % in X / Y / Z, and making the base rigid about 70 / 70 / 35 %. The uprights are only 3.1 L each, so per litre they're the next place to add material (wider or deeper, which also grows the base and the enclosure).
- **Uprights.** Optimizing the base and the uprights together (`analysis/topopt_frame.py`, one shared budget, the uprights free to grow anywhere beside the table under the beam) moves about 3 L from the base into the uprights and turns each upright into a raked buttress: the foot runs about 300 mm forward along the base and the front face slopes up to just ahead of the beam (`analysis/figs/frame_34.6_upright.png`). That alone, at the same total material, is +15 / +7 / +24 % in X / Y / Z. With 4 L less material in total the frame ends up about 20 % less stiff than before, so cutting material isn't worth it.
- The castable version drawn now is a solid full-width buttress (one sloped board in the mould, no cores) with its foot 290 mm ahead of the beam face: 5.3 L per upright (5.5 L since the toe got a 20 mm nose). The base was then re-optimized around the longer feet (`analysis/figs/frame2_base_*.png`): it wants full depth under the feet and a wider tie across the split. Total +8.9 L of epoxy granite (about $75) for +39 / +11 / +35 %.
- The base optimization (`analysis/topopt.py`, pockets open to the bottom) puts the base material into girders under the rails, a tie between them behind the split and knees to the sockets, and takes it out of the perimeter walls, the cross rib at the split and the skin. The castable version drawn here has the same volume as before (28.4 L) and is 5 % stiffer on average (at least 3 % in every load case); the beam change adds the rest.
- The cast-in steel is mostly there for the joints. Replacing the base tubes with solid casting would cost only about 1 %; the beam tubes are worth 6-7 %. Six #6 rebars instead of the beam tubes would be about 3 % less stiff, so the tubes stay.
- Cutting epoxy from the base is cheap in money but expensive in stiffness. At the optimized layout's best: 24.5 L matches the old base, 21 L loses 11 %, and below about 18.5 L (the minimum skin, walls and cover around the steel) it falls off fast. Epoxy granite costs about $8.50/L, so even the best case saves about $35. A free-form base with closed box sections (lost foam cores) is only about 8 % stiffer than open pockets at the same volume, which isn't worth the trouble.

Material and the 8.5 % mix. The model uses E = 30 GPa for the epoxy granite. Published values for well-graded 7-10 % mixes are 25-45 GPa (one commercial mineral casting lists 25 GPa); the university studies use 15-30 % resin with powder fillers and aren't comparable. Granite is about 70 GPa and the epoxy about 3 GPa, so a lower epoxy fraction is as stiff or stiffer as long as every stone is wetted and there are no voids; what it gives up is a little damping (the resin does the damping) and some margin against dry spots. The frame's stiffness scales almost directly with E:

E of the epoxy granite	X	Y	Z
20 GPa	58.0	80.3	72.3
25 GPa	68.3	95.5	84.1
30 GPa (assumed)	**78.2**	**110.1**	**95.5**
35 GPa	87.9	124.4	106.6
40 GPa	97.3	138.4	117.4

So the packing of the mix matters more than the last percent of epoxy. The build guide's jar test and tap-tested bars check both before anything is cast.

Vibration (`analysis/modal.py`). The frame on its four feet, with 12 kg of X carriage, Z axis and spindle on the beam blocks and 12 kg of table, vise and work on the Y blocks:

Mode	Frequency	Motion
1	135 Hz	gantry sways side to side (X); base twisting on its feet carries 39 % of it
2	268 Hz	beam rocks front to back (Y)
3	290 Hz	base bends, front to back
4	343 Hz	gantry sway with the base, second form

Bricks this coarse run a little stiff, so the real frequencies are probably 10-20 % lower. Mode 1 is at the bottom of the tooth-pass band for steel (roughly 130-400 Hz: 4,000-8,000 rpm with 2-3 flutes) and modes 2-4 are inside it. That's normal for a machine this size; the frame is still several times stiffer than the spindle and tool (static X/Y/Z above), and chatter on a mill like this usually comes from the spindle, Z axis and tool. If a frequency rings, change the spindle speed by 10-20 % rather than the frame. A solid bench (the base twisting on its feet is in every mode) and sand in the hollow posts and sockets help more than geometry changes.

Lightening holes in the uprights (`analysis/topopt_upright.py`, `compare_uprights.py`). The uprights were optimized again on their own, with the cuts forced to go straight through (windows) or to stop at a 20 mm web on the inner face (pockets from the outside), keeping 90 % and 80 % of their material (run on the frame before the toe and edge cleanup, which added 1-3 %):

Uprights (both)	Removed	X	Y	Z	Mode 1 / 2	Epoxy saved
Solid, as drawn		77.5	107.5	93.0	135 / 264 Hz	
One window each, 90 %	1.05 L, 2.5 kg	75.0	106.5	91.2	136 / 263 Hz	~$3
One outer pocket each, 90 %	1.05 L	75.6	106.8	92.0	136 / 261 Hz	~$3
Two windows each, 80 %	2.1 L, 4.9 kg	71.7	104.8	88.9	136 / 255 Hz	~$6
Two outer pockets each, 80 %	2.1 L	73.1	105.6	90.3	135 / 254 Hz	~$6

The holes land where you'd expect, in the low-stress middle of the buttress just ahead of the front post (`analysis/figs/upright_*.png`), but they don't pay: 10 % of the upright costs 1-3 % of the frame's stiffness and saves about $3 of epoxy (the 3 gal kit is bought either way), and the natural frequencies don't move because the mass that matters is the beam and the head on top. Each window would also need a printed liner. The uprights stay solid; the frame optimization above already moved material into them, not out.

Figures in `analysis/figs` (and `renders/frame_base_underside.png`): `frame_34.6_*` (base and uprights optimized together), `frame2_base_*` (base around the new uprights), `base_current.png` (the previous base), `base_final.png` (solid depth from the top in each column), `base_pocket_*` and `base_free_*` (optimization results at 18-28.5 L), and `tradeoff.png` (base stiffness against base volume), `upright_*` (lightening holes, at 90 and 80 % of the upright).

Rerun it with a Python that has numpy, scipy, matplotlib and scikit-sparse 0.5 or later (older scikit-sparse falls back to a much slower solver):

```
/Applications/FreeCAD.app/Contents/Resources/bin/freecadcmd analysis/voxelize.py   # ->
analysis/vox_current.*
cd analysis
python baseline.py                                  # stiffness and each module's share (about 1
min)
FEA_VOX=vox_previous sh run_sweep.sh                # base optimizations (about 10 min each)
FEA_VOX=vox_previous python evaluate.py to_*.npz    # table + figures
FEA_VOX=vox_current python whatif.py                # stiffness of the model as drawn
python alumina.py                                   # alumina grit: mix modulus and frame stiffness
python modal.py                                     # natural frequencies (about 30 s)
python topopt_upright.py 0.9 [web]                  # upright holes (about 10 min each)
python compare_uprights.py to_upright_*.npz         # their stiffness and frequencies
```

`vox_current` is whatever `build_cnc.py` draws now (`voxelize.py` rewrites it); `vox_previous` is the frame before these changes, which the optimizations and the "previous design" numbers above use. `VOX_P='{"beam_d": 160}' VOX_OUT=analysis/vox_test` on the voxelize command tries a parameter change.

Chips and coolant

Stage	What	Where in the model
Collect	Sealed base top; gutters between the uprights and the enclosure walls fall toward a trough across the front, which falls to one outlet	`BaseFrontCasting`, `BaseRearCasting`
Protect	Roll-up 0.1 mm stainless strips in front of and behind the table, a spiral spring cover on the rear screw, a drip shield over the beam rails, brush seals where the motors pass through the panels	`WayCovers`, `CoverRoller*`, `DripShield`, `BrushSeal*`
Drain	The trough drops straight down through the front wall to a cast-in brass nipple over the chip basket	`DrainOutlet`
Clean	Slide-out sump under the base, between four 6 in steel legs cast into the corners: a full-size 4 in stainless steam-table pan (~13 L, ~9 L working). Front to back: 1/3-size perforated pan as the chip basket, magnet bar, printed underflow baffle, 100 um screen in a printed frame, pump	`Sump*`, `Leg*`

Stage	What	Where in the model
Return	24 V brushless submersible pump (switched by the controller's M8 output; no mains in the coolant), push-to-connect at the pump, hose under the base and up the back to a bulkhead in the back panel, along the printed cable chain, a loop down to a manifold on the spindle clamp, two Loc-Line nozzles	`ReturnHose`, `Bulkhead`, `CoolantLines`, `CoolantNozzle*`

To empty or clean the sump, pop the hose off the pump and slide the pan out the front (set `sumpOut` in the `Params` sheet to see it pulled out). The legs raise the machine 117 mm and it still sits on four M12 leveling feet, now in the leg ends. Use a water-soluble coolant and keep it topped up; skim tramp oil off the clean end now and then.

Cable chain. One printed chain on top of the beam carries the coolant line, Z motor, spindle power and spindle water hoses to the carriage (17 links, 38 x 21 outer, 20 mm pitch, R38). Print it in PETG, not PLA. The model shows a simplified link; for the real parts, start from a proven printable parametric chain sized to about 30 x 15 inside and print three links to check the fit before printing the rest. The table and Z axes don't need chains: the table carries no cables and Z uses a short service loop.

Parts that change shape with the axes. The way-cover strips, the screw cover, the cable chain and the nozzle hose are drawn for the home position. The collision check and the renders rebuild them at every pose, but jogging `posX` / `posY` / `posZ` in the GUI won't redraw them; re-run the macro to see them elsewhere.

Cost

Checked against real listings (2026-10-09/10, the sourcing list). The list is in three parts: **1, the frame** (castings, steel, rails, ballscrews, chip protection) about **$1,355** (including $150 of alumina grit for the beam and uprights); **2, the electronics and spindle** about **$430** plus the spindle the builder picks ($250-375 for a 1.5 kW 110 V kit, $430-600 for the 2.2 kW kit with a 120 V step-up VFD); **3, the enclosure and flood coolant** about $650-700. The steel and aluminium are priced as remnants and the joint hardware from a hardware store; the remnant prices are estimates. The estimate below was made before that check and is too low; it assumed AliExpress-level prices. It's kept here to show what each cost-down change saves.

Rough estimates in USD at hobby-supplier prices (AliExpress/Amazon-type sources, late 2026). They're not quotes; check current listings. The cost-down column keeps the machine's capability: same travel, rigidity, spindle, closed-loop X/Y, flood coolant and enclosure.

Item	Before	Cost-down	Change
Epoxy + aggregate	$240-315	$200-255	Hollow rear base, 8.5 % epoxy mix (well-graded aggregate + whiting fines), bulk kits
Cast-in steel, rods, dowels, fasteners	$90-140	$90-140	
Rail risers	$60-90	$25-40	Cold-rolled flat bar bedded in epoxy and replicated, instead of MIC6 strips
MIC6 table, carriage, Z plate	$120-180	$120-180	Remnant/drop plate is often much cheaper
Rails + blocks (table, Z)	$110-160	$110-160	Beam rails owned
Ballscrew kits (table, Z)	$70-100	$70-100	Beam kit owned
X/Y closed-loop motors + drivers	$170-220	$140-180	2 Nm 23HS30-E1000: lower inductance, holds torque better at speed, 37 mm shorter

Item	Before	Cost-down	Change
Brackets, couplers	$20-30	$20-30	One bracket owned
Spindle, VFD, clamp, cooling loop	$280-370	$280-370	
Electronics	$150-200	$150-200	
Enclosure	$220-300	$145-210	3 mm ACM walls and roof; polycarbonate only in the door
Coolant system	$105-180	$105-180	
Way covers, seals	$65-105	$65-105	
Lubrication	$40-60	$10-15	Grease gun on the blocks' grease nipples instead of a central pump
Gaskets, inserts, feet	$30-45	$30-45	
Printed cable chain	$3	$3	PETG
Total	**~$1,780-2,500**	**~$1,560-2,210**	

Bigger savings mean giving something up. Options, roughly in order of impact:

- **Phase it.** Build without the enclosure and flood coolant first (mist or air blast). That saves about $250-400 up front. The base still has the gutters and mounting inserts, so both can be added later.
- **Polyester instead of epoxy** for the castings: about half the resin cost, but it shrinks more, so you rely more on the epoxy-replicated mounting surfaces.
- **1.5 kW 110 V spindle** instead of the 2.2 kW you can upgrade to 240 V later: about $40-70 cheaper, less power for steel.
- **Open-loop motors** on X/Y: about $60-80 cheaper, but you lose stall and crash detection.

Reference dimensions

Component	Source	Notes
HGR15 rail, HGH15CA block	HIWIN Linear Guideway catalog G99TE23-2203, HG series (p.38)	Ball-groove positions are not published; estimated.
SFU1204 / SFU1605 nuts, screw end machining	TBI Motion SFU sheet L010974 (via Anaheim Automation)	
BK10/BK12, BF10/BF12 supports	TBI Motion sheets L011175 (BK), L011178 (BF)	BK locknut size estimated. BK12 counterbore depth (1.5) is as printed; worth checking.
DSG16H nut housing	makerstore.cc drawing TR-BS-DSG16-00	
DSG12H nut housing (24 mm bore)	zyltech / vxb listings	Centre height and holes estimated.
NEMA 23 motors	StepperOnline drawings 23HS45-4204D-E1000, 23HS30-5004D-E1000, 23HS30-3004S	
Motor bracket	No published drawing	Sized from kit listings: motor face ~47 mm from the BK.
D25 L30 coupler	Seller listings (5 N·m, 8x10 / 6.35x10 bores)	Hub depth estimated.
GDZ80-2.2 spindle	StepperOnline drawings GDZ80-2.2A/B	Some features scaled from the drawing (±1 mm). Body length varies by seller (213/230/240): measure yours.

Component	Source	Notes
80 mm cast clamp	cnconline.co.za 80 mm spindle bracket drawing	148 W x 80 D, 14 foot, bore centre 50 off the foot.
ER20 nut	DIN 6499 / spindle drawing	Chinese spindles ship a 30 AF hex nut.
2020 extrusion	Misumi HFS5-2020 catalog p.2-511	Owned V-slot is OpenBuilds profile (6.25 throat).

Before casting anything, measure the parts you actually have, especially the owned ANWOK rails, blocks and SFU1204 kit. Clone parts can differ from these catalogs by a millimetre or two.